NLP exam

Apprentice Model

Project goal

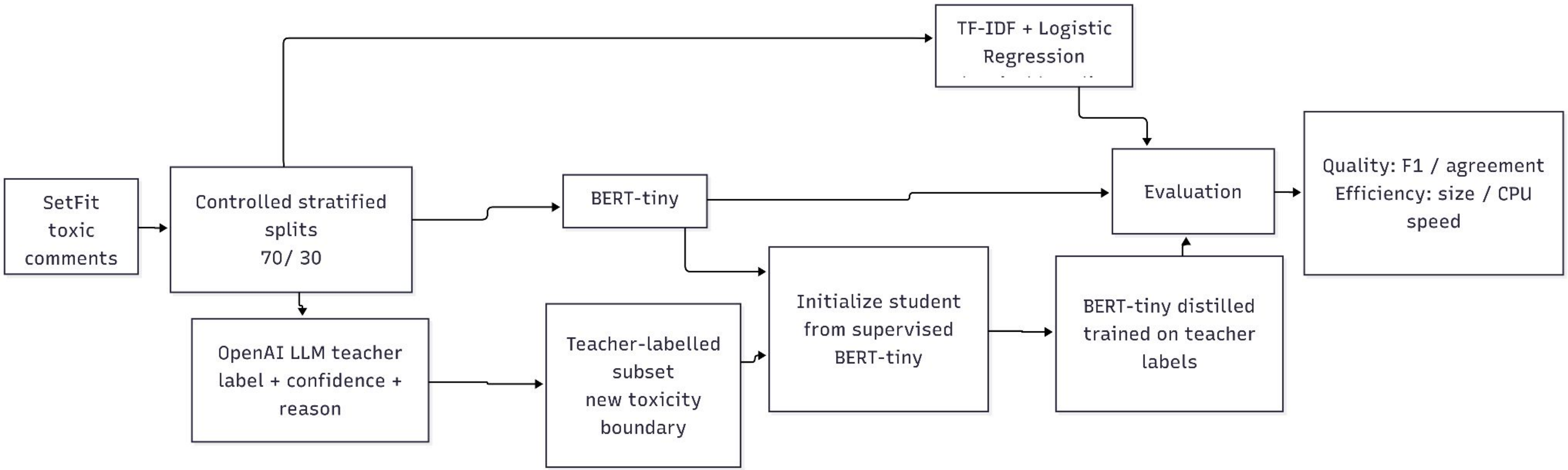
Apprentice Model:
train a small BERT-tiny student to imitate an LLM teacher for toxic comment classification

1. Quality
Compare TF-IDF, supervised BERT-tiny, and distilled BERT-tiny.

2. Efficiency
Measure model size and CPU inference speed.

3. Teacher imitation
Study whether the student learns the teacher’s toxicity boundary.

Pipeline


TF-IDF + Logistic Regression
SetFit toxic comments
Controlled stratified splits 70/ 30
BERT-tiny
Evaluation
Quality: F1 / agreement Efficiency: size / CPU speed
Initialize student from supervised BERT-tiny
BERT-tiny distilled trained on teacher labels
OpenAI LLM teacher label + confidence + reason
Teacher-labelled subset new toxicity boundary

Teacher labelling changes the decision boundary

Split	Rows	Origins toxic, %	Teacher toxic, %	Agreement
Train	1500	30	45.6	76.27
Validation	300	30	46.7	77.33
Test	300	30	44.3	80.33

Results and efficiency

Distilled Student Partially Imitates the Teacher

Model	Evaluation target	F1	Size, Mb	Time per example, ms
TF-IDF + Logistic Regression	Original labels	0.63	0.89	0.06
BERT-tiny supervised	Original labels	0.66	17.42	1.61
BERT-tiny distilled	Teacher labels	0.69	17.42	1.59

Practical motivation and conclusions

Why Distillation Is Useful

Model	Deployment	Time per example, ms	Throughput, ex/s
OpenAI LLM teacher	Remote API	5440	0.18
Distilled BERT-tiny	Local CPU	1.56	642

Future work

Better Imitation and Safer Evaluation

1. Soft-label distillation

Use teacher probabilities with KL-divergence instead of only hard labels.

2, Threshold calibration

Tune the decision threshold to better match the teacher’s toxic/non-toxic distribution.

3. Larger teacher-labelled set

Reduce variance and improve generalization.

4. Bias and explainability checks

Use minimal-pair tests, SHAP, saliency, or CAVs to verify that the student learns toxicity, not identity or topic shortcuts.